Group Representation Theory

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1 Representations

Definition 1.0.1. A **representation** of a group G is a pair (V, ρ) where V is a vector space and $\rho : G \to GL(V)$ is a group homomorphism. So a left group action $G \curvearrowright V$ by invertible linear transformations. Much like with group actions, a representation is called **faithful** if ρ is injective.

Definition 1.0.2. For a group G, a homomorphism of representations $T : (V, \rho) \to (V', \rho')$ is a linear map $T : V \to V'$ such that for all $T \circ \rho(g) = \rho'(g) \circ T$ for all $g \in G$.

If we think about this categorically, then the category of k-representations of a group G is $\mathbf{Vect}_k^G$.

From now on, unless specified otherwise, all representations will be of a finite group G on a finite dimensional complex vector space. This lets us identify $GL(V)$ with $GL_n(\mathbb{C})$, the normal dictionary between linear maps and matrices from linear algebra applies everywhere.

Lemma 1.0.3. Two finite-dimensional representations (V, ρ) and (V', ρ') are isomorphic iff $\dim(V) = \dim(V')$ and there exists bases B and B' respectively such that there exists some $P \in GL_n(\mathbb{C})$ such that

$$[\rho'(g)]_{B'} = P^{-1}[\rho(g)]_B P$$

for all $g \in G$. In this case, the **matrix representations** are called **equivalent**.

We can now consider a few examples of representations.

For any group G, we can consider the one-dimensional representation $\rho(g) = \mathrm{id}_{\mathbb{C}}$ for all g, this is called the **trivial respresentation**.

If ζ is a nth root of unit, then $C_n = \langle g \rangle$ has a one-dimensional representation by multiplication by $\rho(g^k)(z) = \zeta^k z$ for all $k \in \mathbb{Z}$, when $n = 1$ this recovers the trivial respresentation.

Let $G = S_n$, and let each $\sigma \in S_n$ act as the corresponding matrix of a basis B, i.e. $\rho(\sigma)(b_i) = b_{\sigma(i)}$ for all $b_i \in B$.

If $G = S_n$, then there is a one-dimensional represpresentation with $\rho(g) = (\mathrm{sign}(g))$. This is equal to applying the determinant to the premutation respresentation.

Let $G = D_{2n}$, this has natural two-dimensional representation on $\mathbb{R}^2$ sending each g to the associated transformation.

For any group action $G \curvearrowright X$ on a set X, there is an associated action of G on $\mathbb{C}[X]$, when we take $X = G$, and the action is by left multiplication, then the action of G on the $\mathbb{C}$-algebra $\mathbb{C}[G]$ is called the **regular representation**. For any representation (V, ρ), the natrual map:

$$\phi : \mathrm{Hom}_G(\mathbb{C}[G], V) \to V \qquad \phi(T) = T(e)$$

is a G-linear isomorphism, with inverse $\phi^{-1}(v)(g) = \rho(g)v$.

1.1 Subrepresentations

Definition 1.1.1. A **suprepresentation** of a representation (V, ρ) is a subspace $W \leq V$ such that $\rho(g)(W) \subseteq W$ for all $g \in G$, we also call W a G-invariant subspace. Subrepresentations are called **proper** and **nonzero** if W satisfies the respective condition from linear algebra.

When W is finite dimensional, as $\rho(g)|_W \in GL(W)$ is a bijectiction, we in fact have $\rho(g)(W) = W$. When W is infinite dimensional this need not be true, only when $\rho(g)(W) \subseteq W$ *for all* $g \in G$ do we get $\rho(g)(W) = W$ for all $g \in G$.

Definition 1.1.2. A representation (V, ρ) is **irreducible** (or **simple**) if it is nonzero, and it does not admit any proper nonzero subrepresentations. Otherwise, it is called **reducible**.

Proposition 1.1.3. *If G is a finite group with (V, ρ) an irreducible representation, then V is finite dimensional.*

Proof. Let $v \in V$ be nonzero, and write W for the span of the $\rho(g)(v)$ for all $g \in G$, then this is a subrepresentation of V, as for all $h \in G$:

$$\rho(h) \sum_{g \in G} a_g \rho(g) v = \sum_{g \in G} a_g \rho(hg) v \in W$$

for all $a_g \in \mathbb{C}$. As v is nonzero, so is W, so by irreducibility of (V, ρ), $V = W$, finite dimensional. □

For an alternate proof, consider the map $\mathbb{C}[G] \to V$ induced by some nonzero $v \in V$, the image of this is a finite dimensional non-zero subrepresentation.

Proposition 1.1.4. *If $T : (V, \rho) \to (V', \rho')$ is a homomorphism of representations, then $\ker(T)$ and $\mathrm{im}(T)$ are subrepresentations of V and V' respectively.*

Proof. $T(v) = 0 \implies T(gv) = g(Tv) = 0$ for all $g \in G$, $gT(v) = T(gv) \in \mathrm{im}(T)$ for all $g \in G$. □

Definition 1.1.5. A **quotient representation** of a repressentation (V, ρ_V) by a subrepresentation (W, ρ_W) is the vector space V/W with action $\rho(g)(v + W) = \rho(g)v + W$.

Proposition 1.1.6. *For $T : (V, \rho) \to (V', \rho')$ is a homomorphism of representations, there is an isomorphism of representations:*

$$(V, \rho)/\ker(T) \cong \mathrm{im}(T)$$

Proof. We just have to show the map from the first isomorphism theorem $\overline{T}$ is G-linear. For all $g \in G$ and $v \in V$:

$$\overline{T} \circ \rho(g)(v + \ker(T)) = T \circ \rho(g)(v) = \rho(g)T(v) = \rho(g)\overline{T}(v + \ker(T))$$ □

Recall the following fact from linear algebra:

Lemma 1.1.7. For $T \in \mathrm{End}(V)$ with $T^2 = T$, there is the direct sum decomposition $V = \ker(T) \oplus \mathrm{im}(T)$.

Proof. Send $v \in V$ to $(v - T(v), T(v))$, this is immediately linear and bijective. $\square$

In this case, we call T a **projection operator**.

Corollary 1.1.8. If $T \in \mathrm{End}(V)$ is a G-linear projection operator, then the direct sum of the previous lemma is a direct sum of subrepresentations.

Definition 1.1.9. A nonzero representation is **decomposible** if it can be written as a direct sum of two proper nonzero subrepresentations, it is called **indecomposable** otherwise.

A representation that is a direct sum of irreducible subrepresentations is called **semisimple**.

Indecomposable representations are semisimple iff they are irreducible. There are still examples of indecomposable but not irreducible representations:

Let $G = \mathbb{Z}$ and consider the two-dimensional representation:

$$\rho(n) = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

This has the x-axis as a subrepresentation, but this is the only common one-dimensional eigen-space of $\rho(G)$, so there is not a full decomposition into irreducible subrepresentations.

1.2 Maschke's theorem

Definition 1.2.1. For (V, ρ) a representation, and $W \leq V$ a subrepresentation, a **complementary subrepresentation** is a subrepresentation $U \leq V$ such that $V = W \oplus U$.

Theorem 1.2.2 (Maschke)**.** Let (V, ρ) be a finite dimensional representation of a finite group G. For any subrepresentation $W \leq V$, there exists a complementary representation.

Proof. We want to construct a projection operator with W as its image. By linear algebra, we can find a U such that $V = W \oplus U$, and the map $T(v) = T(w + u) = w$ is a good candidate. Unfortunately, this isn't necessarily G-linear, so we have to modify it, have $\tilde{T} : V \to V$ define as:

$$\tilde{T} = |G|^{-1} \sum_{g \in G} \rho(g) \circ T \circ \rho(g)^{-1}$$

we first have to show this linear map is G-linear, for all $h \in G$ consider:

$$\tilde{T} \circ \rho(h) = |G|^{-1} \sum_{g \in G} \rho(g) \circ T \circ \rho(g)^{-1} \rho(h) = |G|^{-1} \sum_{g' = h^{-1} g \in G} \rho(h)\rho(g') \circ T \circ \rho(g')^{-1} = \rho(h) \circ \tilde{T}$$

so T' is G-linear. To prove it is a projection operator we must show $\tilde{T}|_W = \mathrm{id}$, for all $w \in W$:

$$\tilde{T}(w) = |G|^{-1} \sum_{g \in G} \rho(g) \circ T \circ \rho(g)^{-1}(w) = |G|^{-1} \sum_{g \in G} \rho(g)\rho(g)^{-1}(w) = |G|^{-1} \sum_{g \in G} w = w$$

Finally, we must show $\mathrm{im}(\tilde{T}) = W$, by the previous statement we know $\in (\tilde{T}) \subseteq W$, but as T is a projection operator on W and W is G-invariant, we also have $W \subseteq \mathrm{im}(\tilde{T})$. $\square$

We can now do induction on the dimension of the irreducible subspace we consider to get:

Corollary 1.2.3. Any finite dimensional representation of a finite group can be written as a direct sum of irreducible subrepresentations. i.e. V is semisimple.

Corollary 1.2.4. Finitie dimensional indecomposable representations of finite groups are irreducible.

This doesn't actually require the vector space to be finite dimensional. Note, we haven't said anything about this decomposition being unique, there are examples where unqiueness does and doesn't hold:

If $G = 1$, then representations are the same as vector spaces, so the only irreducible representation of G is $\mathbb{C}$. For any bigger representation, the decomposition of is exactly the decomposition of the vector space, which is exactly a basis.

Let $A = GL_n(\mathbb{C})$ be diagonal, with distinct diagonal entries which are mth roots of unity for some $m \geq n$. Then consider the n-dimensional representation of $C_m = \langle g \rangle$ with $\rho(g) = A$. Then I claim the decomposition into subrepresentations is exactly $\mathbb{C}^n = \mathbb{C}e_1 \oplus \cdots \oplus \mathbb{C}e_n$.

Let $G = C_3 = \{1, g, g^2\}$, and $\mathbb{C}[G]$ be the regular representation, then multiplication by g has matrix:

$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & \zeta^2 & \zeta \\ 1 & \zeta & \zeta^2 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \zeta & 0 \\ 0 & 0 & \zeta^2 \end{pmatrix} P^{-1}$$

so we are in teh case of the previous example, and so this representation splits as the sum of three one-dimensional representations. This example can be generalise to any C_n, natrually

1.3 Schur's lemma

Theorem 1.3.1 (Schur's lemma)**.** Let (V, ρ_V) and (W, ρ_W) be irreducible representations of G:

(i) If $T : V \to W$ is a G-linera map, then either T is an isomorphism or the zero map.

(ii) Suppose V is finite-dimensional, if $T : V \to V$ is G-linear, then $T = \lambda I$ for some $\lambda \in \mathbb{C}$.

Proof. (i) Apply the first isomrophism theorem, and consider the possibilities for $\ker(T)$ and $\operatorname{im}(T)$

(ii) By linear algebra, every nonzero T is a nonzero eigenvector v such that $T(v) = \lambda v$, now $T - \lambda I$ is a non-injective map, so by the previous part $T = \lambda I$. □

The rotation representation of C_n on $\mathbb{C}^2$, where the generator is a counter-clockwise rotation by $2\pi/n$, as every rotation is G-linear and these aren't always scalar multiples of the identity. When we consider this on $\mathbb{R}^2$ our matrices no longer have real eigenvalues for $n \geq 3$ so there are no real 1D subspaces.

1.4 Representations of abelian groups

Proposition 1.4.1. If G is an arbitrary abelian group G, then every finite-dimensional irreducible representations of G is one-dimensional.

Proof. Let (V, ρ_V) be the representations, then for all $g \in G$, $\rho_V(g)$ is G-linear and so a multiple of identity matrix, any 1D subspace is now a subrepresentation. □

From the proof we should realise $g \in G$ arbitrary is central iff $\rho(g)$ is a scalar matrix for all irreducible representations ρ. We need G to be finite to get the converse as we need Maschke's theorem to go from $\rho(g)$ G-linear on all irreducible representations to all representations.

If, in the proposition, V isn't finite, then consider the action of $\mathbb{C}(x)^\times$ on $\mathbb{C}(x)$ by multiplication, then V is irreducible and infinite-dimensional and G is abelian.

Corollary 1.4.2. For $G = C_n$ there are exactly n irreducible representations given by $\rho_i(g) = \zeta_n^i$.

So the irreducible representation of C_n on $\mathbb{C}^2$ from earlier can be seen as $\rho_1 \oplus \rho_{-1}$. We will generalise this to the representation of the product of groups later.

All 1D representations of a finite group G is conjugate to a homomorphism $\rho : G \to \mathbb{C}^\times$, which factors through $G/[G,G]$, so by our discussion of products the number of 1D representations is $|G/[G,G]|$, we can actually use this to deduce the size of the abelianisation of certain groups, S_n and D_{2n} are the easiest ones to start.

1.5 Homomorphisms

Definition 1.5.1. If V and W are representations of G over k, then $\operatorname{Hom}_G(V, W)$ is the vector space of G-linear maps $V \to W$, write $\operatorname{End}_G(V)$ for $\operatorname{Hom}_G(V, V)$.

If V and W are irreducible, Schur's lemma can now be equivalently stated as

$$\dim \operatorname{Hom}_G(V, W) = \begin{cases} 1 & \text{if } V \cong W \\ 0 & \text{otherwise} \end{cases}$$

Definition 1.5.2. If (V, ρ_V) and (W, ρ_W) are representations of G, then the map

$$\rho_{\mathrm{Hom}(V,W)}(g)(\phi) = \rho_W(g) \circ \phi \circ \rho_V(g)^{-1}$$

is a representation. Write $V^G = \{v \in V \mid \forall g \in G,\ \rho_V(g)(v) = v\}$ for the G-invariant subspace of the representation.

Proposition 1.5.3. $\mathrm{Hom}_G(V, W) = \mathrm{Hom}(V, W)^G$.

Proof. $T \in \mathrm{Hom}_G(V, W)$ iff $\forall g \in G$, $T\rho_V(g) = \rho_W(g)T$ iff $T = \rho_W(g)T\rho_V(g) = \rho_{\mathrm{Hom}(V,W)}(g)(T)$, iff $T \in \mathrm{Hom}(V, W)^G$. □

The proof of Maschke's theorem was forming a projection operator $V \to V^G$.

1.6 Decompositions

By Maschke's theorem, if G is a finite group and (V, ρ) is a finite-dimensional representation. We can write $V \cong V_1^{r_1} \oplus \cdots \oplus V_m^{r_m}$ as a direct sum of irreducible subrepresentations where the V_i are pairwise nonisomorphic.

Proposition 1.6.1. $r_i = \dim \mathrm{Hom}_G(V_i, V) = \dim \mathrm{Hom}_G(V, V_i)$.

Proof. As Hom commutes with $\oplus$, we just divide up into subrepresentations and use our new form of Schur's lemma. □

Corollary 1.6.2. The decomposition of V into irreducible are unique up to isomorphism of the V_is.

Proof. Each irreducible representation V_i, up to isomorphism, occurs exactly r_i times in V. □

Corollary 1.6.3. Given such a decompositions, we have $\dim \mathrm{End}_G(V) = r_1^2 + \cdots r_m^2$.

Now remembering the formula $\mathrm{Hom}_G(\mathbb{C}[G], V) \cong V$, we get

Corollary 1.6.4. If V has the above decomposition, then we have the two statements

$$\mathbb{C}[G] \cong V_1^{\dim V_1} \oplus \cdots \oplus V_m^{\dim V_m} \implies |G| = \sum_{i=1}^{m} (\dim V_i)^2$$

We can use this arithmetic to reliably find all the irreducible representations of a finite group. We call such a set of all irreducible nonisomorphic representations a **full set**.

Definition 1.6.5. The **adjoint representation** $(\mathbb{C}[G], \rho_{\mathrm{ad}})$ is given by

$$\rho_{\mathrm{ad}}(g)\left(\sum_{h \in G} a_h h\right) = \sum_{h \in H} a_h g h g^{-1}$$

Theorem 1.6.6. If G is a finite group and $(V_1, \rho_1), \ldots, (V_m, \rho_m)$ is a full set of irreducible representations, then m is the number of conjugacy classes in G.

Proof. First consider the linear map $\Phi = (\rho_1, \ldots, \rho_m) : \mathbb{C}[G] \to \mathrm{End}(V_1) \oplus \cdots \oplus \mathrm{End}(V_m)$, as these have equal dimension we just need to check Φ is injective. If $f \in \mathbb{C}[G]$ satisfies $\rho_i(f) = 0$ for all i, then by Maschke's theorem, for every finite dimensional representation (W, ρ_W) of G—a sum of V_is—will have $\rho_W(f) = 0$. If we apply this to $\mathbb{C}[G]$, then $\rho_{\mathbb{C}[G]}(f)(1) = f = 0$.

Now, for all $g, h \in G$, we have $\rho_i(ghg^{-1}) = \rho_{\mathrm{End}(V_i)}(g)\rho_i(h)$, so Φ is an isomorphism of representations from $(\mathbb{C}[G], \rho_{\mathrm{ad}})$.

Finally, we show that $(\mathbb{C}[G], \rho_{\mathrm{ad}})^G$ have a basis $f_1, \ldots, f_{m'}$ where f_j is a sum of all the elements in one of the m' conjugacy classes of G, if we have $f = gfg^{-1} \in \mathbb{C}[G]$, then the coefficients of f across a conjugacy class are all equal, so this set spans, and is obviously LI.

We can now take the dimension of the induced linear isomorphism

$$\mathbb{C}^{m'} \cong (\mathbb{C}[G], \rho_{\mathrm{ad}})^G \cong \bigoplus_{i=1}^{m} \mathrm{End}_G(V_i) \cong \mathbb{C}^m$$ □

1.7 Duals and tensors

We now for a finite dimensional vector space V there is a noncanonical isomorphism $V \cong V^* := \mathrm{Hom}(V, \mathbb{C})$, if V is a representation then by letting G act trivially on $\mathbb{C}$ we get the dual representation of V^*: $\rho_{V^*}(g)(f)(v) := (f \circ \rho_V(g)^{-1})(c)$. On 1D representation show this is just inverting g.

Proposition 1.7.1. There is a canonical G-linear inclusion $V \hookrightarrow (V^*)^*$ which is an isomorphism if V is finite-dimensional.

Proof. We know the linear algebra part, so we just have to check G-linearity which looks like

$$\rho_{(V^*)^*}(g)(\phi_v)(f) = \phi_v(\rho_{V^*}(g)^{-1}(f)) = \phi_v(f\rho_V(g)) = f(\rho_V(g)(v)) = \phi_{\rho_V(g)(v)}(f) \qquad \square$$

Definition 1.7.2. Given vector space V, W, their **tensor product** if $V \otimes W$, the quotient of $\mathbb{C}[V \times W]$ by the subspace spanned by

$$a(v,w) - (av,w), \quad a(v,w) - (v,aw), \quad (v+v',w) - (v,w) - (v',w), \quad (v,w+w') - (v,w) - (v,w')$$

where we write the image of (v,w) under the quotient as $v \otimes w$.

Proposition 1.7.3. Given vector spaces U, V, W, every bilinear map $F : V \times W \to U$ factors through the map $V \times W \to V \otimes W$.

Proof. We get a linear map $\mathbb{C}[V \times W] \to U$ by extending F, as F is bilinear it is zero on the kernel and by the first isomorphism theorem we have a factorisation. $\square$

Corollary 1.7.4. We have the following natrual isomorphisms for arbitrary vector spaces U, V, W

$$\begin{aligned} V \otimes W &\cong W \otimes V & (v \otimes w) &\mapsto (w \otimes v) \\ (U \otimes V) \otimes W &\cong U \otimes (V \otimes W) & ((u \otimes v) \otimes w) &\mapsto (u \otimes (v \otimes w)) \\ U \otimes (V \oplus W) &\cong (U \oplus V) \otimes (U \oplus W) & u \otimes (v,w) &\mapsto (u \otimes v, u \otimes w) \\ \mathrm{Hom}(U \otimes V, W) &\cong \mathrm{Hom}(U, \mathrm{Hom}(V,W)) & [u \otimes v \mapsto w] &\mapsto [u \mapsto [v \mapsto w]] \end{aligned}$$

and an injective linear map

$$V^* \otimes W^* \to (V \otimes W)^* \qquad f \otimes g \mapsto [v \otimes w \mapsto f(v)g(w)]$$

which is an isomorphism is V is finite-dimensional.

This says the categorical data $(\mathbf{fVect}_{\mathbb{C}}, \otimes, \mathbb{C}, \oplus, 0)$ is a categorification of the semiring $(\mathbb{N}, \times, 1, +, 0)$. This should invoke lots of question about what certain canonical and noncanonical isomorphims of vector spaces mean in terms of arithmetic.

Definition 1.7.5. If (V, ρ_V) and (W, ρ_W) are representations, then we can form the **tensor product representation**

$$\rho_{V \otimes W}(g)(v \otimes w) = \rho_V(g)(v) \otimes \rho_W(g)(w)$$

We want to know how to construct tensor products of matrices, this is called the **Kronecker product**. If M and N are $n \times n$ and $m \times m$ complex matrices respectively, then their tensor product is the $mn \times mn$ matrix $(M \otimes N)_{ij,kl} = M_{ik}N_{jl}$ where the ijth element of the basis for $\mathbb{C}^n \otimes \mathbb{C}^m$ is $e_i \otimes e_j$, so

$$(M \otimes N)(e_i \otimes e_j) = \sum_{k=1}^{n} \sum_{l=1}^{m} (M_{ik}e_k) \otimes (N_{jl}e_j)$$

Definition 1.7.6. If G and H are groups with representations (V, ρ_V) and (W, ρ_W) respectively, we form their **exernel tensor product** on $V \boxtimes W := V \otimes W$ with $\rho_{V \boxtimes W} : G \times H \to GL(V \otimes W)$ by $\rho_{V \boxtimes W}(g,h) = \rho_V(g) \otimes \rho_W(g)$.

If $\Delta : G \to G \times G$ is the diagonal map, then $\rho_{V \otimes W} = \rho_{V \boxtimes W} \circ \Delta$.

Proposition 1.7.7. If G and H are finite groups, then the irreducible representations of $G \times H$ are precisely the externel tensor product of irreducible representations of G and H respectively.

Proof. First, consider $\mathrm{End}_{G\times H}(V\boxtimes W) = \mathrm{End}(V\boxtimes W)^{G\times H} = \mathrm{End}_G(V\boxtimes W)^H$ and considser the map $\mathrm{End}(W)\to\mathrm{End}_G(V\boxtimes W)$ by $T\mapsto I\otimes T$, this is clearly H-linear and injective, to show it is surjective we can use dimensions: as G-representations, $V\boxtimes W\cong V^{\dim W}$, so

$$\dim\mathrm{End}_G(V\boxtimes W) = \dim\mathrm{End}_G(V^{\dim W}) = \sum_{\dim W}\dim\mathrm{Hom}_G(V,V^{\dim W}) = (\dim W)^2 = \dim\mathrm{End}(W)$$

where by Schur's lemma we now have

$$\mathrm{End}_{G\times H}(V\boxtimes W) = \mathrm{End}_G(V\boxtimes W)^H = \mathrm{End}(W)^H = \mathrm{End}_H(W) = \mathbb{C}$$

so $V\boxtimes W$ is in fact irreducible.

To show these are all the irreducible representations, we can just count

$$|G\times H| = |G|\,|H| = \sum_{i=1}^{n}(\dim V_i)^2\sum_{j=1}^{m}(\dim W_j)^2 = \sum_{ij}\dim(V_i\otimes W_j)^2$$

where V_i and W_j are the irreducible represenations of G and H respectively. □

Proposition 1.7.8. If V, V' and irreducible representations of G and W, W' are irreducible representations of W, then $V\boxtimes W\cong V'\boxtimes W'$ iff $V\cong V'$ and $W\cong W'$.

Proof. If $V\boxtimes W\cong V'\boxtimes W'$, then by the above argument we have $V^{\dim W}\cong V'^{\dim W'}$, and thus $V\cong V'$ and so $W\cong W'$ □

These proofs both uses bases, if we give a basis free argument, then we can also loose our assumption that G and H be finite.

Proof. Once again, consider the map $\mathrm{End}(W)\to\mathrm{End}_G(V\boxtimes W)$, and for each $w\in W$ and $f\in W^*$ consider the G-linear composition $V\xrightarrow{I\otimes w}V\otimes W\xrightarrow{T}V\otimes W\xrightarrow{I\otimes f}V$, which must be a multiple of the identity, so for some $v\in V$, there exists a $w'\in W$ such that $T(v\otimes w) = v\otimes w'$. As V is irreducible all other $v'\in V$ are in the G-span of v, so we can write

$$T(v'\otimes w) = T\circ\sum_{g\in G}(a_g\rho(g)\otimes I)(v\otimes w) = \sum_{g\in G}(a_g\rho(g)\otimes I)(v\otimes w') = (v'\otimes w')$$

so $T(v'\otimes w) = v'\otimes w'$ is in the image of $\mathrm{End}(W)$, so our map is surjective.

Now, given an irreducible representation U of $G\times H$, let V be any irreducible G-representation isomorphic to a G-subrepresentation of U, then we have a map $f: V\boxtimes\mathrm{Hom}_G(V,U)\to U$ by $(v\otimes\phi)\mapsto\phi(v)$ where $\mathrm{Hom}_G(V,U)$ is a H-representation. Now as U is irreducible, and f is nonzero, it must be surjective. Let $W'\leq\mathrm{Hom}_G(V,U)$ the largest subspace with $V\boxtimes W'$ in the kernel of f, then we have a surjection $V\boxtimes(\mathrm{Hom}_G(V,U)/W')\to U$ and any irerducible subrepresentation $W\leq\mathrm{Hom}_G(V,U)/W'$, which exists by the assumption of finite dimensionality, the induced $V\boxtimes W\to U$ must also be nonzero hence surjective.

So if U is any irreducible representation of $V\boxtimes W$ for V, W irreducible, and we take $V'\boxtimes W'\to U$ the above surjection, the nonzero composite $V'\boxtimes W'\to V\boxtimes W$, by the next paragraph, must have $V\cong V'$ and $W\cong W'$, so we can set them equal and the composition becomes $V\boxtimes W\to V\boxtimes W$ nonzero, so a multiple of the identity with image U.

If $V\boxtimes W\cong V'\boxtimes W'$, then the first paragraph can be generalised to show $\mathrm{Hom}_G(V\boxtimes W, V'\boxtimes W') = \mathrm{Hom}_G(V,V')\otimes\mathrm{Hom}(W,W')$, which by Schur's lemma is nonzeroo iff $V\cong V'$ as G-representations, we then do the same for $W\cong W'$. □

We once again use $W = V = \mathbb{C}(x)$ a $\mathbb{C}(x)^\times$-representation as a counterexample, as $\mathbb{C}(x)\otimes\mathbb{C}(x)$ isn't irreducible, as the multiplication map $\mathbb{C}(x)\otimes\mathbb{C}(x)\to\mathbb{C}(x)$ is a surjective representation with $f\otimes 1 - 1\otimes f$ in the kernel for every $f\in\mathbb{C}(x)$, with $f\otimes 1 - 1\otimes f$ nonzero as it lies outside the kernel of the multiplication map $\mathbb{C}(x)\otimes\mathbb{C}(y)\to\mathbb{C}(x,y)$.

2 Characters

Definition 2.0.1. For (V, ρ_V) a finite-dimensional representation of a group G, the **character** $\chi_{(V,\rho_V)} : G \to \mathbb{C}$ is the function $\chi_{(V,\rho_V)}(g) = \operatorname{tr} \rho_V(g)$.

We know this is well-defined as the trace of a matrix is invariant under conjugacy. We will now work to show that the character is a full invariant of finite-dimensional representations of finite groups up to isomorphism.

Lemma 2.0.2. Equivalent matrix representations, and isomorphic abstract representations, have the same character.

For all representatiosn (V, ρ_V) of G, as $\rho(1) = 1$, we have $\chi_V(1) = \dim(V)$, so χ_V is multiplicative iff $\dim(V) = 1$.

Proposition 2.0.3. Every character χ_V of a group G is constant on conjugacy classes.

Proof. For all $g, h \in G$, $\chi_V(h^{-1}gh) = \operatorname{tr} \rho_V(h^{-1}gh) = \operatorname{tr}(\rho_V(h)^{-1}\rho_V(g)\rho_V(h)) = \operatorname{tr}(\rho_V(g)) = \chi_V(g)$. □

Functions from groups which are constant on conjugacy classes are called **class functions**, the set of class functions $f : G \to \mathbb{C}$ is a subspace of the vectorspace $\operatorname{Fun}(G, \mathbb{C})$, called $\operatorname{Fun}_{\text{cl}}(G, \mathbb{C})$. When G is finite, $\dim \operatorname{Fun}_{\text{cl}}(G, \mathbb{C})$ is the number of conjugacy classes in G. If we give $\operatorname{Fun}(G, \mathbb{C})$ the adjoint action $(g \cdot f)(h) = f(g^{-1}hg)$, then $\operatorname{Fun}_{\text{cl}}(G, \mathbb{C}) = \operatorname{Fun}(G, \mathbb{C})^G$.

Proposition 2.0.4. Let (V, ρ_V) be a finite-dimensional representation of G, then for all $g \in G$ with finite order, then $\chi_V(g^{-1}) = \overline{\chi_V(g)}$ and $|\chi_V(g)| \leq \dim V$ with identity holding iff $\rho_V(g)$ is a scalar multiple of the identity.

Proof. If g has finite order $g^n = e$, then $\rho_V(g)^n = I$, and thus its characteristic polynomial divides $x^n - 1$ which has distinct eigenvalues, so $\chi_V(g)$ is a sum of eigenvalues, the two results follow immediately. □

2.1 Characters of sums and products

Proposition 2.1.1. Let (V, ρ_V) and (W, ρ_W) be finite-dimensional representations of a group G, then $\chi_{V \oplus W} = \chi_V + \chi_W$ and $\chi_{V \otimes W} = \chi_V \chi_W$. If, G is also finite, then $\chi_{V^*} = \overline{\chi_V}$ and $\chi_{\operatorname{Hom}(V,W)} = \overline{\chi_V}\chi_W$.

Proof. Let $\mathcal{B} = \{v_1, \dots, v_m\}$ and $\mathcal{C} = \{w_1, \dots, w_n\}$ be bases of v and W respectively, fix $g \in G$, and let $M := [\rho_V(g)]_{\mathcal{B}}$ and $N := [\rho(g)]_{\mathcal{C}}$, so $\chi_V(g) = \operatorname{tr} M$ and $\chi_W(g) = \operatorname{tr} N$.

Let $\mathcal{D} = \mathcal{B} \sqcup \mathcal{C}$ be a basis for $V \oplus W$, then $[\rho_{V \oplus W}(g)]_{\mathcal{D}}$ is the block diagonal $\operatorname{diag}(M, N)$, so $\operatorname{tr} \rho_{V \oplus W}(g) = \operatorname{tr} \rho_V(g) + \operatorname{tr} \rho_W(g)$.

Now let $\mathcal{B} \otimes \mathcal{C}$ be the basis for $V \otimes W$, then $[\rho_{V \otimes W}(g)]_{\mathcal{B} \otimes \mathcal{C}} = M \otimes N$, which has trace

$$\operatorname{tr}(M \otimes N) = \sum_{i,j} (M \otimes N)_{ij,ij} = \sum_{ij} M_{ii} N_{jj} = \operatorname{tr}(M) \operatorname{tr}(N)$$

Let $\mathcal{E}$ be the dual basis $\{f_1, \dots, f_m\}$ of $\mathcal{B}$ for V^*, then we have already computed $[\rho_{V^*}(g)]_{\mathcal{E}} = (M^{-1})^T$, and $\operatorname{tr}(M^{-1})^T = \operatorname{tr} M^{-1} = \operatorname{tr}(\rho_V(g^{-1}))$, we already know $\chi_V(g^{-1}) = \overline{\chi_V(g)}$.

We know $\operatorname{Hom}(V, W) \cong V^* \otimes W$, so this part is already done. □

If we have $G = \mathbb{Z}$ and $a \in \mathbb{C}^\times$ and consider the 1D represnetaiont $\rho(m) = a^m$, then $\chi_\rho(-1) = a^{-1} \neq \overline{a}$ in general. So the last two parts actually only require all the elements of G be of finite order.

2.2 Inner products of characters

Definition 2.2.1. For two functions $f_1, f_2 : G \to \mathbb{C}$, their **inner product** is

$$\langle f_1, f_2 \rangle := \frac{1}{|G|} \sum_{g \in G} f_1(g) \overline{f_2(g)}$$

Proposition 2.2.2. This is a Hermitian inner product on $\operatorname{Fun}(G, \mathbb{C})$.

Proof. Sesquilinearity, conjugate-symmetry and positive definiteness can all be read off the definition. □

We quickly proof the following seemingly-unrelated lemmata.

Lemma 2.2.3. Let $S : V \to V$ be a linear projection, then $\dim \operatorname{im} S = \operatorname{tr} S$.

Proof. Choose a basis $\mathcal{B}$ for V specialised to $V = \ker S \oplus \operatorname{im} S$, then $[S]_{\mathcal{B}} = \operatorname{diag}(0_{\dim \ker S}, I_{\dim \operatorname{im} W})$. $\square$

Lemma 2.2.4. If (V, ρ_V) is a representation of a group G, then there is an isomorphism of vector spaces $\operatorname{Hom}_G(\mathbb{C}_{\text{triv}}, V) \cong V^G$, by $\phi \mapsto \phi(1)$.

Proof. This map is a linear isomorphism $\operatorname{Hom}(\mathbb{C}, V) \cong V$, and as G acts trivially on $\mathbb{C}$, this is G-linear, so we can take G-invariants and get the result. $\square$

Theorem 2.2.5. Let (V, ρ_V) and (W, ρ_W) be finite-dimensional representations of a finite group G, then $\langle \chi_V, \chi_W \rangle = \dim \operatorname{Hom}_G(V, W)$.

Proof. First, we do the case where $V = \mathbb{C}_{\text{triv}}$, recall the G-linear projection $S : W \to W^G$ by $S = |G|^{-1} \sum_{g \in G} \rho(g)$, then we have

$$\langle \chi_{\mathbb{C}_{\text{triv}}}, \chi_W \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_W(g)} = \frac{1}{|G|} \sum_{g \in G} \operatorname{tr}\left(\overline{\rho_W(g)}\right) = \operatorname{tr} \frac{1}{|G|} \sum_{g \in S} \overline{\rho_W(g)} = \operatorname{tr} \overline{S} = \overline{\operatorname{tr} S}$$

which, by the previous two lemmata, is $\overline{\dim W^G} = \dim W^G = \dim \operatorname{Hom}_G(\mathbb{C}_{\text{triv}}, W)$.

We now just observe $\langle \chi_V, \chi_W \rangle = \left\langle \chi_{\mathbb{C}_{\text{triv}}}, \chi_{\operatorname{Hom}(V,W)} \right\rangle = \dim \operatorname{Hom}_G(V, W)$. $\square$

Corollary 2.2.6. If V and W are irreducible representations, then $\langle \chi_V, \chi_W \rangle = 1$ iff $V \cong W$ and is 0 otherwise. Or, for a full set of irreducible representations V_i of G, the χ_{V_i} form an orthonormal basis for $\operatorname{Fun}_{\text{cl}}(G, \mathbb{C})$.

We can also rephrase all of our statements about dimensions and decompositions from earlier in terms of characters, like

$$V \cong \bigoplus_{i=1}^{m} V_i^{\langle \chi_{V_i}, \chi_V \rangle} \quad \chi_V = \sum_{i=1}^{n} \langle \chi_{V_i}, \chi_V \rangle \chi_{V_i} \quad \langle \chi_V, \chi_V \rangle = \sum_i r_i^2 = \sum_i \langle \chi_{V_i}, \chi_V \rangle = 1 \Leftrightarrow V \text{ irreducible}$$

So there are two natural bases for $\operatorname{Fun}_{\text{cl}}(G, \mathbb{C})$, either the Kronecker delta functions on conjugacy classes of G, or irreducible characters. These are both orthogonal, but we have to scale the Kronecker delta functions by the size of the class to make them orthonormal.

In the case $G = \mathbb{Z}/n\mathbb{Z}$, the relationship between the Kronecker delta basis, and the irreducible characters is exactly the discrete Fourier transform, so for finitei abelian groups we are just taking the direct sum of these transformations. The general correspondence between these two bases for general G can be thought of as a nonabelian generalisation of the Fourier transform.

2.3 Character tables

Definition 2.3.1. For a finite group G with $V_1, \ldots, V_m$ a full set of irreducible representation, and $\mathcal{C}_1, \ldots, \mathcal{C}_m$ the conjugacy classes with $g_i \in \mathcal{C}_i$ a representative, the **character table** of G is the table whose (i, j)th entry is $\chi_{V_i}(g_j)$.

2.4 Kernels of representatins

2.5 Automorphisms

3 Algebras

3.1 Modules

3.2 Semisimple algebras

3.3 Characters

3.4 Centers